

Probabilistic Models of the Visual Cortex

Fall 2025 Homework 4

Prof. Alan Yuille

Instructions

1. **Deadline:** Dec 14, 2025 at 23:59:59
2. **Penalty for late submission:** 20% score deduction per day (or part thereof)
3. **Submission format:** PDF file named `<your_jhed>-hw4.pdf`.
4. **Submission platform:** Gradescope
5. **Warning:** Start the coding part early!
6. **Collaboration:** Feel free to discuss the questions with friends, but be sure to write your answers entirely independently! Do not discuss coding questions.
7. **Questions?** Start a Canvas Discussion, attend office hours, or send an email!

Question 1: Deep Networks (26 points)

1. Describe the basic ideas and the structure of a convolutional neural network.
You may want to sketch it too! (5 points)
2. What is the main difference between multi-layer perceptrons and convolutional neural networks? (3 points)
3. How do the convolutional layers help CNNs perform better in an image classification task? (3 points)
4. What is max-pooling? (2 points)
5. What is average-pooling? (2 points)
6. Why do we need pooling in deep networks? (3 points)
7. What is the potential problem of using a sigmoid activation function in a deep convolutional network? (3 points)
8. What is a ReLU function and how does it help solve the problem? (5 points)

Question 2: Performance Measures and Adversarial Examiners (20 points)

1. Name five ways that deep networks are not robust. (5 points)
2. How does lack of robustness relate to dataset bias and the combinatorial complexity of images? (5 points)
3. What are the limitations of current performance measures? (5 points)

4. What are the basic ideas of adversarial examiners? Discuss the trade-offs between adversarial examiners and testing by random sampling. (5 points)

Question 3: Compositional Models and Theory (10 points)

1. What are the potential advantages of compositional models? What are some current drawbacks? (5 points)
2. How can compositional models be used to defend against occluders and patch attacks? (5 points)

Question 4: Transformers (15 points)

1. Describe the basic building block of transformer networks. What is self-attention? What are the queries, keys, and values? (5 points)
2. Do transformers rely on a regular input grid, like convolutional neural networks? Why are transformers particularly suitable for tasks like learning by masked-image-modelling (where some parts of the image are masked out)? (5 points)
3. What is cross-attention? How can cross-attention be formulated as k-means clustering (in some cases)? (5 points)

Question 5: 3D Geometry of the World (15 points)

1. What is perspective projection? What are its properties (e.g., what are the images of parallel lines; what are vanishing points)? What is orthographic projection? In what circumstances is it a good approximation to perspective projection? (5 points)
2. How can velocity be estimated if the optical flow is known correctly? In which situation can singular value decomposition (SVD) be used to estimate shape? (5 points)
3. How can we estimate depth by egomotion if most of the viewed scene is unchanging (e.g., driving down a road). How can we estimate objects that are moving relative to the scene? How can we use this to estimate depth and train a deep network for estimating depth from a static image? Under what conditions would you trust this deep network? (5 points)

Question 6: Vision Question and Answering (20 points)

1. What are referring expressions? What are questions-and-answers? How do these relate to Visual Turing tests? (5 points)
2. What was the first neural network strategy for combining vision and language? How interpretable was this approach? (5 points)
3. What are neuro-modular networks? Why are they suited to referring expressions and questions-and-answers? To what extent are they interpretable? (5 points)

points)

4. Are neuro-modular methods particularly good for generalizing to out-of-distribution situations? Can they be improved by training the vision and language components independently? (5 points)

Question 7: Experimental Section: Neural Networks (20 points)

In this question, you will build your own neural network to solve a classic classification problem. Here's a link to the Jupyter notebook with all the code you'll need: https://github.com/Tiezheng11/ProbabilisticModelsOfVisualCognition2025FA/blob/main/homework_4/neural_networks.ipynb

For this homework, you do NOT need to submit any code to us. We only expect to see generated plots and answers to the questions in the notebook.